

Midpoint Report

CS 510:Explorations in Data Science

Objective:

Conduct exploratory data analysis using visualization by applying principles of data science. Learn Tableau tool software & python programming language to create interactive data visualizations. I will use travel time, weather & incident data sets found on portal <http://portal.its.pdx.edu/> for visualizing/analysing the impacts of crashes & severe weather on travel times.

Approach:

The three phases of the project approach are:

1. Data understanding & Preparation: This involves initial data collection, understanding data requirements & cleansing /formatting data.
2. Exploratory Analysis/Modelling: Develop methodology, determine important variables, build models, assess models.
3. Visualization & Presentation: Communicate results, determine best method/graph to present insights based on analysis.

Team Size :1

Update on progress with respect to project milestones:

1. Understand the topic, Research the topic & Gather data.

Identified the objective of the analysis & conducted research by reading journal articles & research papers .Collected data from the portal <http://portal.its.pdx.edu/> .This includes data related to

incidents, travel time & weather. Identified required variables for my analysis & performed data cleansing/formatting .

2. Prepare & present research presentation.

Studied research papers & prepared a presentation for the class on July 16.

3. Develop methodology & select a software.

Learned tableau tool software & working on it.

4. Create visualizations & interpret the results.

Started creating charts & graphs suitable for the requirement. Examples of some of the plots created are: weather conditions vs Average travel time (weather conditions are analysed throughout the day along with the travel time readings), No. of crashes vs Travel time, Crashes per county , Crashes per route, Crashes vs Travel time vs No. of lanes affected (visualized by dual axis graphs) etc.

5. Final project presentation.

Final project presentation is scheduled on August 6th.

Are you where you thought you'd be at this point?

Yes, I am on track with the project and will be completing the project on schedule.

Date/time of scheduled midpoint meeting with the prof.

Tuesday , July 21st, 3:00 PM.

07/23/2020